

LAB 4:-

1. Round Robin CPU scheduling

```
#include <stdio.h>

int main() {
    int n, tq;
    printf("Enter number of processes: ");
    scanf("%d",&n);

    int at[n], bt[n], rt[n], ct[n], tat[n], wt[n], rtm[n];
    int started[n];

    printf("Enter Arrival Time and Burst Time:\n");
    for(int i=0;i<n;i++){
        printf("P%d: ", i+1);
        scanf("%d %d",&at[i],&bt[i]);
        rt[i] = bt[i];
        started[i] = 0;
    }

    printf("Enter Time Quantum: ");
    scanf("%d",&tq);

    int time = 0, completed = 0;

    while(completed < n){
        for(int i=0;i<n;i++){
            if(rt[i] > 0 && at[i] <= time){

                if(started[i] == 0){
                    rtm[i] = time - at[i];
                    started[i] = 1;
                }

                if(rt[i] > tq){
                    time += tq;
                    rt[i] -= tq;
                }
                else{
                    time += rt[i];
                    ct[i] = time;
                    rt[i] = 0;
                    completed++;
                }
            }
        }
    }
}
```

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    }
}

float avg_wt = 0, avg_tat = 0, avg_rt = 0;

printf("\nAT\tBT\tCT\tTAT\tWT\tRT\n");

for(int i=0;i<n;i++){
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];

    avg_wt += wt[i];
    avg_tat += tat[i];
    avg_rt += rtm[i];

    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        at[i], bt[i], ct[i], tat[i], wt[i], rtm[i]);
}

avg_wt /= n;
avg_tat /= n;
avg_rt /= n;

printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f", avg_tat);
printf("\nAverage Response Time = %.2f\n", avg_rt);

return 0;
}

```

OUTPUT:

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Enter number of processes: 4
Enter Arrival Time and Burst Time:
P1: 0 4
P2: 0 5
P3: 0 3
P4: 0 2
Enter Time Quantum: 2

AT      BT      CT      TAT      WT      RT
0       4       10     10       6       0
0       5       14     14       9       2
0       3       13     13      10       4
0       2       8      8       6       6

Average Waiting Time = 7.75
Average Turnaround Time = 11.25
Average Response Time = 3.00

```

2. Multilevel Queue Scheduling

```
#include <stdio.h>
```

```
int main() {

    int n;
    printf("Enter number of processes: ");
    scanf("%d",&n);

    int pid[n], at[n], bt[n], type[n];
    int ct[n], tat[n], wt[n], rt[n];

    int gantt[100], gtime[100];
    int g = 0;

    printf("Enter PID AT BT TYPE(0-System,1-User):\n");

    for(int i=0;i<n;i++)
        scanf("%d %d %d %d",&pid[i],&at[i],&bt[i],&type[i]);

    int time = 0;

    for(int i=0;i<n;i++){
        if(type[i] == 0){

            if(time < at[i])
                time = at[i];

            gantt[g] = pid[i];
            gtime[g] = time;
            g++;

            rt[i] = time - at[i]; // Response Time

            time += bt[i];
            ct[i] = time;

            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];
        }
    }

    for(int i=0;i<n;i++){
        if(type[i] == 1){
```

```

        if(time < at[i])
            time = at[i];

        gantt[g] = pid[i];
        gtime[g] = time;
        g++;

        rt[i] = time - at[i];

        time += bt[i];
        ct[i] = time;

        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];
    }
}

gtime[g] = time;

float avg_wt=0, avg_tat=0, avg_rt=0;

printf("\nPID\tAT\tBT\tTYPE\tCT\tTAT\tWT\tRT\n");

for(int i=0;i<n;i++){
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],type[i],ct[i],tat[i],wt[i],rt[i]);

    avg_wt += wt[i];
    avg_tat += tat[i];
    avg_rt += rt[i];
}

avg_wt /= n;
avg_tat /= n;
avg_rt /= n;

printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f", avg_tat);
printf("\nAverage Response Time = %.2f\n", avg_rt);

printf("\nGantt Chart:\n");

for(int i=0;i<g;i++)

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        printf("| P%d ", gantt[i]);
printf("\n");

for(int i=0;i<=g;i++)
    printf("%d  ", gtime[i]);

printf("\n");

return 0;
}

```

OUTPUT:

```

Enter number of processes: 3
Enter PID AT BT TYPE(0-System,1-User):
1 0 4 1
2 0 4 0
2 2 6 0

PID      AT      BT      TYPE    CT      TAT     WT      RT
1         0        4        1       14      14      10      10
2         0        4        0        4        4        0        0
2         2        6        0       10        8        2        2

Average Waiting Time = 4.00
Average Turnaround Time = 8.67
Average Response Time = 4.00

Gantt Chart:
| P2 | P2 | P1 |
0   4   10  14

Process returned 0 (0x0)   execution time : 58.219 s
Press any key to continue.
|

```